

MADISON, MN, USA

WMO: 727466

Lat: 44.986N

Lon: 96.178W

Elev: 330

StdP: 97.43

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-22.4	-28.7	0.3	-23.9	-26.4	0.4	-21.7	14.4	-9.6	13.0	-8.0	3.0	310	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.7	23.3	31.3	22.9	29.1	21.8	25.9	30.1	24.7	29.1	23.6	27.7	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.8	28.1	23.0	18.4	27.2	22.3	17.7	26.4	82.1	31.2	76.6	29.2	72.0	27.4	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.2	9.9	-28.5	35.8	3.6	1.9	-31.1	37.2	-33.2	38.3	-35.2	39.4	-37.8	40.8		
(5)			-28.8	27.6	3.5	1.9	-31.3	29.0	-33.3	30.0	-35.3	31.1	-37.8	32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-9.1	-8.3	-0.8	7.7	14.6	20.2	22.6	20.6	16.6	8.9	1.4	-6.8		
	DBStd	12.62	7.50	7.03	7.06	5.72	4.95	3.77	3.31	3.32	4.79	5.28	6.25	6.81		
	HDD10.0	2439	591	514	338	108	14	0	0	0	5	86	264	519		
	HDD18.3	4360	849	747	592	321	137	22	6	15	88	296	508	778		
	CDD10.0	1484	0	0	4	38	157	307	389	330	202	51	6	0		
	CDD18.3	364	0	0	0	2	21	79	137	87	36	3	0	0		
	CDH23.3	3747	0	0	1	42	282	818	1369	781	407	48	1	0		
	CDH26.7	1273	0	0	0	11	102	283	500	236	131	11	0	0		
(14) Wind	WSAvg	4.2	4.5	4.7	4.8	5.2	5.0	4.0	2.9	2.9	3.6	4.2	4.5	4.6		
(15) Precipitation	PrecAvg	664	18	18	35	66	77	103	86	76	73	64	31	18		
	PrecMax	850	61	43	85	176	123	223	159	160	206	200	89	51		
	PrecMin	480	1	3	10	17	14	33	21	16	16	6	1	1		
	PrecStd	98	15	11	19	36	34	52	39	32	45	50	27	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	11.2	21.2	27.7	33.0	34.0	34.9	33.0	32.8	28.0	21.3	10.9		
		MCWB	4.4	6.5	12.0	16.6	19.1	22.2	26.3	24.7	21.6	15.7	12.4	5.6		
	2%	DB	4.9	7.1	16.2	23.7	29.1	32.2	32.6	31.3	30.2	23.9	17.3	7.0		
		MCWB	2.4	3.8	10.6	13.5	18.3	21.7	25.2	24.1	20.9	14.9	10.8	3.5		
	5%	DB	2.8	3.8	12.3	21.0	26.4	29.8	31.2	29.0	27.7	21.1	13.6	3.8		
		MCWB	1.3	1.8	8.5	12.1	17.0	21.6	24.5	22.8	19.3	13.7	8.1	1.7		
	10%	DB	2.0	2.2	8.6	17.6	23.7	27.7	29.1	27.5	25.9	17.7	10.9	2.4		
		MCWB	0.8	0.8	6.3	10.6	15.6	20.0	22.9	21.5	18.4	11.8	6.9	1.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	6.6	14.0	17.5	22.2	25.7	28.7	27.2	24.2	19.2	14.5	6.6		
		MCDB	8.1	9.7	18.1	24.8	28.6	30.8	32.3	30.4	29.0	24.2	19.5	9.1		
	2%	WB	2.9	3.8	11.1	15.0	20.1	24.2	26.7	25.2	22.9	16.5	11.0	3.5		
		MCDB	4.5	6.4	15.5	21.7	26.3	29.2	30.4	29.8	27.5	21.9	16.0	5.8		
	5%	WB	1.6	2.0	8.4	13.1	18.4	22.8	25.3	23.9	21.5	14.4	8.8	2.2		
		MCDB	3.0	3.8	11.7	19.0	24.3	27.8	30.0	27.9	25.8	19.1	13.1	3.8		
	10%	WB	0.3	0.5	5.9	11.5	16.9	21.5	24.1	22.6	19.6	12.6	6.7	1.0		
		MCDB	1.6	1.8	8.8	16.4	22.1	26.2	27.9	26.2	24.1	16.9	10.2	2.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.0	9.5	11.5	11.8	11.2	11.5	11.8	13.0	11.6	10.5	8.9		
		MCDBR	11.2	11.8	14.8	17.7	16.0	14.1	12.6	13.4	15.6	16.6	15.9	11.5		
	5% WB	MCWBR	9.3	9.4	9.9	9.4	7.8	6.9	7.0	7.4	7.6	8.7	10.0	9.0		
		MCWBR	10.3	11.4	14.0	14.6	13.6	11.6	11.5	12.0	12.6	13.4	14.7	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.278	0.311	0.359	0.378	0.389	0.395	0.400	0.360	0.319	0.284	0.266		
	taud		2.441	2.421	2.429	2.335	2.351	2.378	2.349	2.314	2.410	2.504	2.517	2.454		
	Ebn at Noon		864	917	926	901	891	879	868	850	860	857	838	826		
	Edh at Noon		70	86	99	119	121	118	120	120	100	79	65	62		
(44) All-Sky Solar Radiation	RadAvg		1.80	2.87	3.80	4.78	5.50	6.14	6.58	5.46	4.23	2.74	1.80	1.33		
	RadStd		0.20	0.32	0.37	0.42	0.40	0.40	0.24	0.46	0.31	0.36	0.17	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page